



ADJOINT-ACCELERATED INVERSE MODELLING OF ACOUSTIC CROSS-TALK IN PIEZOELECTRIC INKJET PRINTHEADS

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We develop an adjoint-accelerated inverse modelling framework to generate accurate, interpretable models of the internal fluid dynamics in drop-on-demand piezoelectric inkjet printheads. From Laser Doppler Vibrometry (LDV) measurements of meniscus oscillations, we jointly infer internal acoustic flow parameters and reconstruct the acoustic flow field within the printhead. Despite the sparsity of the data, the inferred model reproduces the observed dynamics with high accuracy. The reconstruction further reveals cross-talk mechanisms not directly accessible experimentally, including direct acoustic excitation between adjacent channels, acoustic wave propagation through the printhead structure, and coupling via the shared ink manifold. Building on these insights, we derive reduced-order models (ROMs) describing the internal printhead dynamics. We then use Bayesian inference for parameter estimation, uncertainty quantification, and model selection. The resulting ROM is both interpretable and predictive, capturing the essential dynamics with a small number of physical parameters. Finally, we demonstrate accurate extrapolation to unseen printheads, highlighting the robustness of the approach. This work establishes a PDE-constrained inverse modelling approach for complex acoustic channel dynamics in industrial applications, enabling predictive simulation and reducing reliance on trial-and-error experimental testing.

References

- [1] Adjoint-based optimization of open-loop control for microfluidics of an inkjet printhead (Javier Lorente-Macías; Matthew P. Juniper), *The Journal of the Acoustical Society of America*.
- [2] Fundamental Fluid Dynamics Challenges in Inkjet Printing (Detlef Lohse), *Annual Review of Fluid Mechanics*.